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SUPERSEDING
NAVFAC DM-38.3
JULY 1981

MILITARY HANDBOOK

YARD CRAFT

AMSC N/A

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ABSTRACT

This handbook contains criteria for the design and construction of nonself-propelled yard craft of the types under the technical cognizance of the Naval Facilities Engineering Command. Yard craft are custom-designed to perform specific functions. The criteria presented in this handbook are common to most types. Information pertaining to design factors, load capacities, and mechanical and electrical systems are also included.

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FOREWORD

This military handbook has been developed from an evaluation of facilities in the shore establishment, from surveys of the availability of new materials and construction methods, and from selection of the best design practices of the Naval Facilities Engineering Command (NAVFACENGCOM), other Government agencies, and the private sector. It uses to the maximum extent feasible, national professional society, association, and institute standards. Deviations from this criteria, in the planning, engineering, design, and construction of Naval shore facilities, cannot be made without prior approval of NAVFACENGCOMHQ Code 04.

Design cannot remain static any more than can the functions it serves or the technologies it uses. Accordingly, recommendations for improvement are encouraged and should be furnished to Commander, Western Division, Naval Facilities Engineering Command, Code 406, P. O. Box 727, San Bruno, CA 94066; telephone (415) 877-7422.

THIS HANDBOOK SHALL NOT BE USED AS A REFERENCE DOCUMENT FOR PROCUREMENT OF FACILITIES CONSTRUCTION. IT IS TO BE USED IN THE PURCHASE OF FACILITIES ENGINEERING STUDIES AND DESIGN (FINAL PLANS, SPECIFICATIONS, AND COST ESTIMATES). DO NOT REFERENCE IT IN MILITARY OR FEDERAL SPECIFICATIONS OR OTHER PROCUREMENT DOCUMENTS.

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WEIGHT HAULING EQUIPMENT AND SERVICE CRAFT DESIGN MANUALS

Criteria Manual	Title	PA
DM-38.01	Weight Handling Equipment	HDQTR
DM-38.02	Dredging Equipment	SOUTHDIV
MIL-HDBK-1038/3	Yard Craft	WESTDIV
DM-38.4	Pile Driving Equipment	LANTDIV
MIL-HDBK-1038/5	Towing Nonself-propelled Floating Structures	WESTDIV

NOTE: Design Manuals, when revised, will be converted to military handbooks and listed in the military handbook section of NAVFAC P-34.

This handbook is issued to provide immediate guidance to the user. However, it may or may not conform to format requirements of MIL-HDBK-1006/3 and will be corrected on the next update.

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TOWING NONSELF-PROPELLED FLOATING STRUCTURES

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Section 1: INTRODUCTION

1.1 Scope. This handbook provides policy and detailed procedures for the design and construction of yard craft under the technical cognizance of the Naval Facilities Engineering Command (NAVFACENGCOM). The design process is presented in sufficient detail to provide the naval procurement officer or engineering manager with the normal performance and construction features as well as military standards and certifications that must be considered in developing U.S. Navy (USN) Yard Craft. Detail design data is left to the individual design team to develop.

1.2 Cancellations. This handbook cancels and supersedes NAVFAC DM-38.3, July 1981.

1.3 References. A number enclosed with ( ) refers to references at the end of this handbook.

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Section 2: GENERAL

2.1 Scope of Craft Covered. This handbook applies to yard craft such as barges, pontoons, lighters and other specialty craft. Each type is generally custom designed and built for a specific purpose. The yard craft referenced in this handbook are nonself-propelled.

2.2 Examples of Navy Yard Craft. Examples of current or potential USN yard craft are illustrated in Table 1 and Figures 1 through 5 of this handbook, Figure 15 of NAVFAC DM-38.2, Dredging Equipment, and Appendix A of NAVFAC P-401, Pontoon System Manual (28).

TABLE 1
Types of Yard Craft

TYPE	CARGO OR EQUIPMENT	REMARKS	FIG.[1]
Open Deck Cargo	Open-deck-holds - convert to covered by covering over	Dry bulk cargo. Available in single-skin, double-bottom or double-skin construction	1
Covered Deck Cargo Barge		Coamings are required for bulk cargo. Packaged type cargo shall be lashed to deck	
Utility Barge	On deck	General Purposes. Symmetrical ends.	
Harbor Lighter	On deck	Sized from 80x30x9 ft to 112x34x11 ft Heavy scantlings incorperated to withstand quick maneuvering forces.	3
Dump Scow	Below-deck Compartments	Use bucket-type dredges or refuse disposal. Use side dump scows in shallow water. Bottom-dump scow for gen. use in deep water. Self-dumping needs deep water at dredging & dumping site.	15 of DM-38.02
Fuel	Below-deck	Single-skin centerline bulkhead and double-skin barge w/out centerline bulkhead are most commonly used.	4
NL Pontoon Bridge	On deck	Constructed of small pontoon units linked to form a barge. Used for cargo or equipment[2].	5

[1]These figures illustrate typical examples of barge construction. They are not intended to indicate actual service craft used by the Navy.

[2]Refer to NAVFAC P-401, Pontoon System Manual, for information and details

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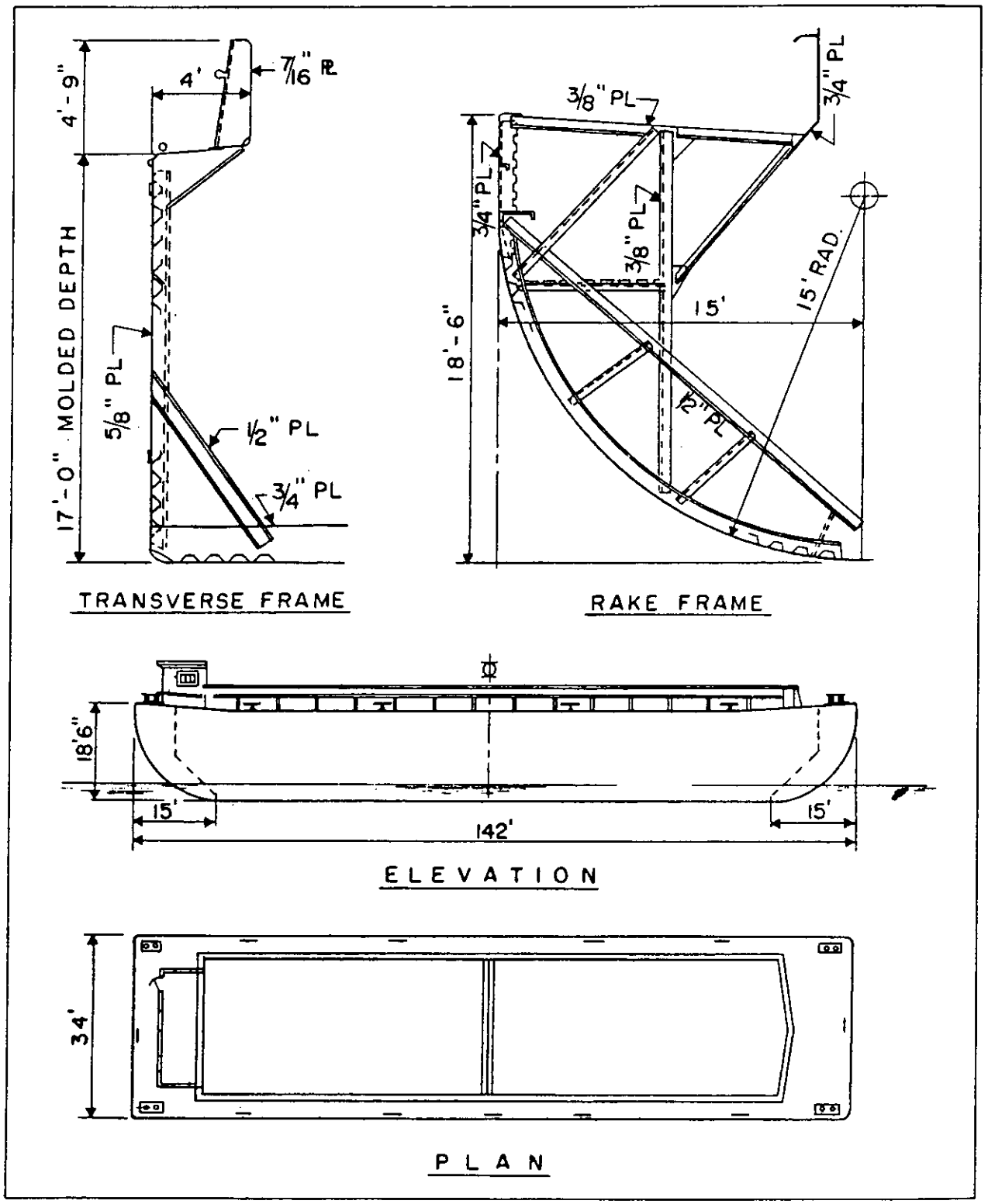


FIGURE 1
Open Deck Barge

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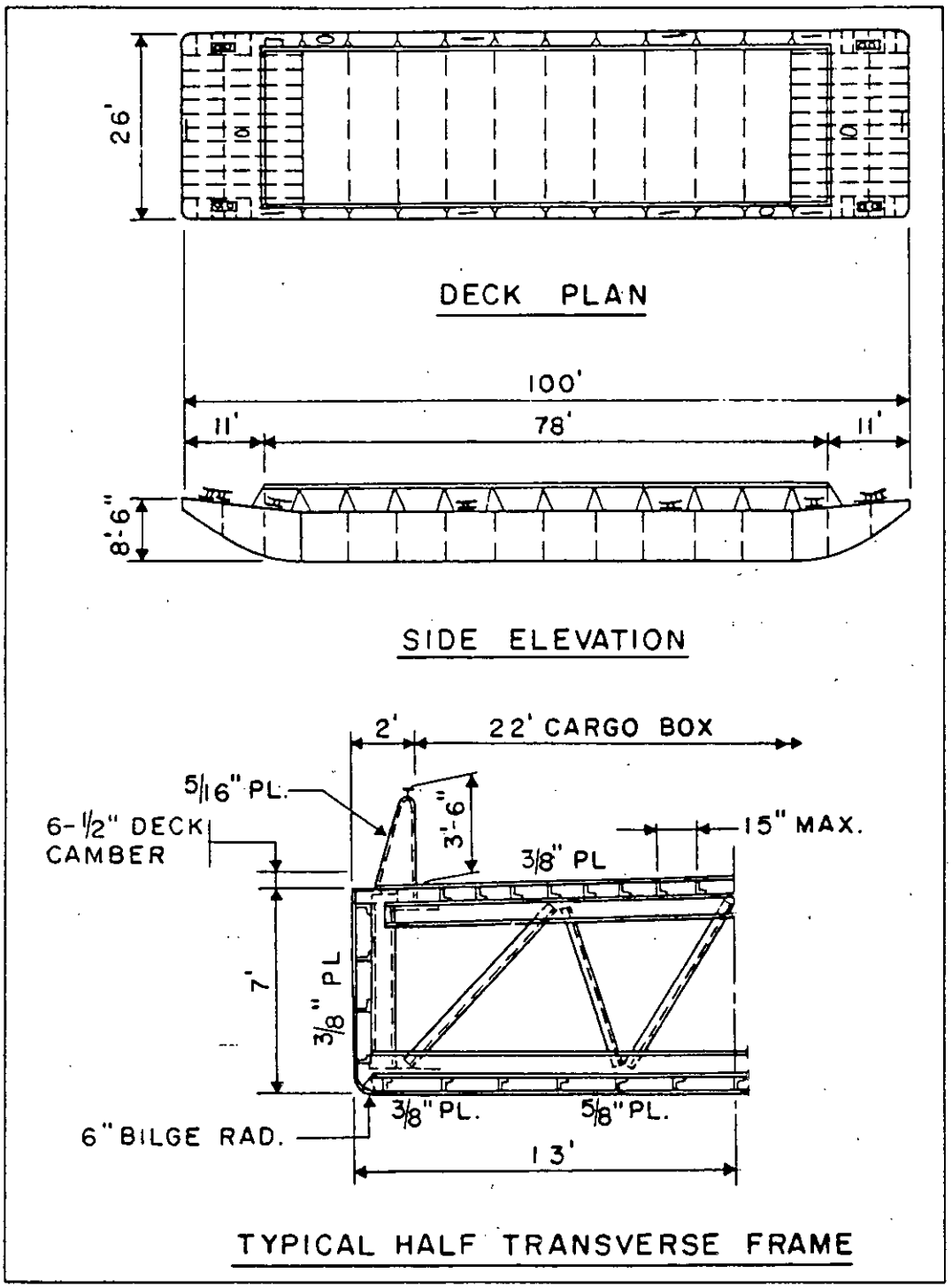


FIGURE 2
Utility Barge

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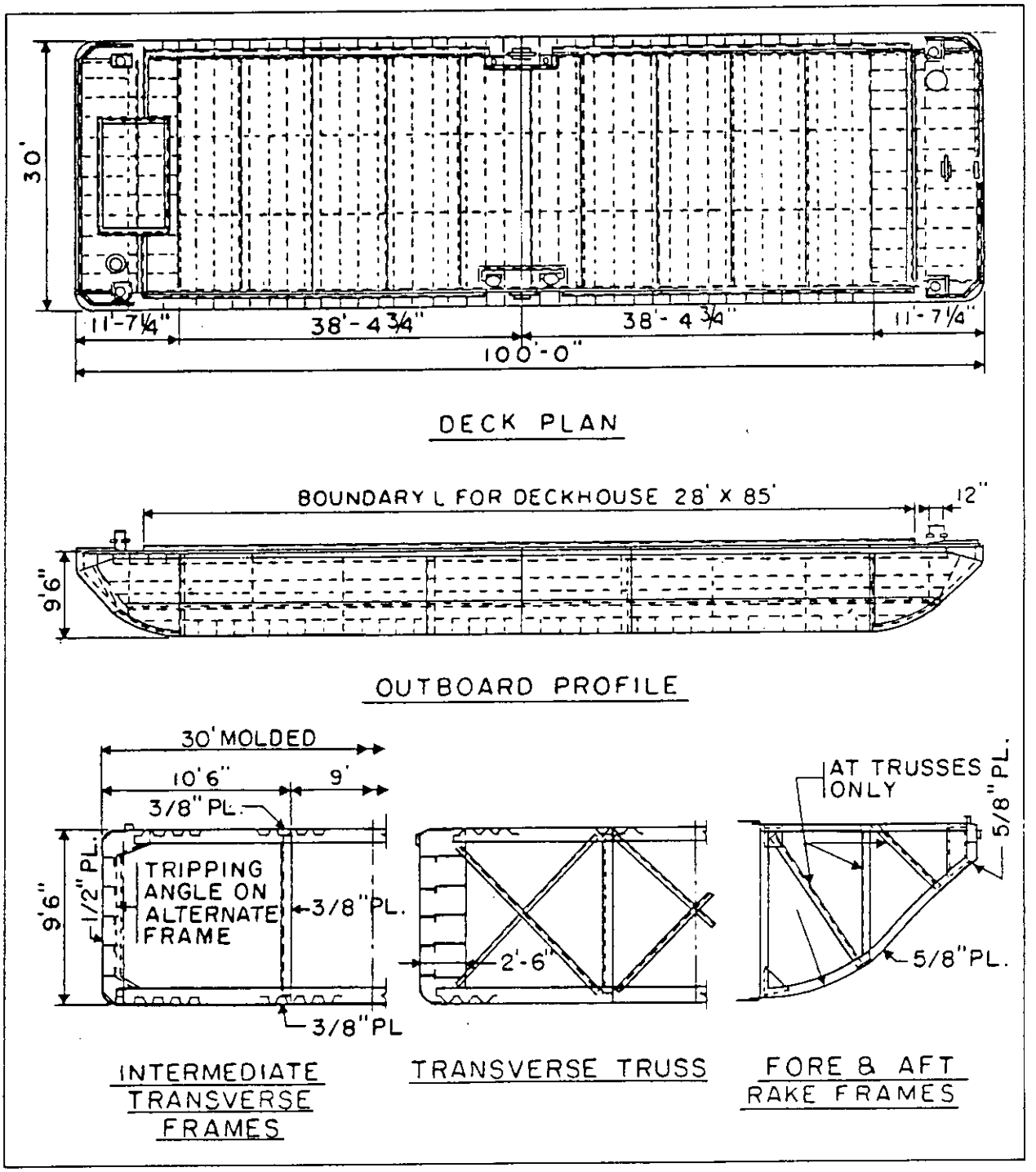


FIGURE 3
Harbor Lighter

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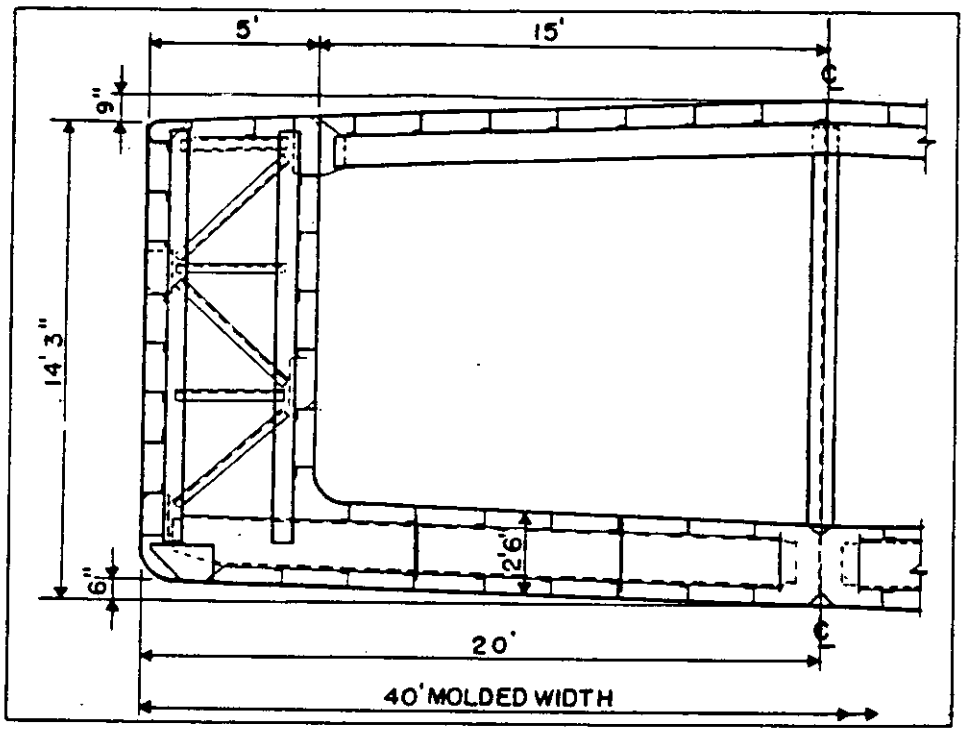


FIGURE 4
Fuel Barge

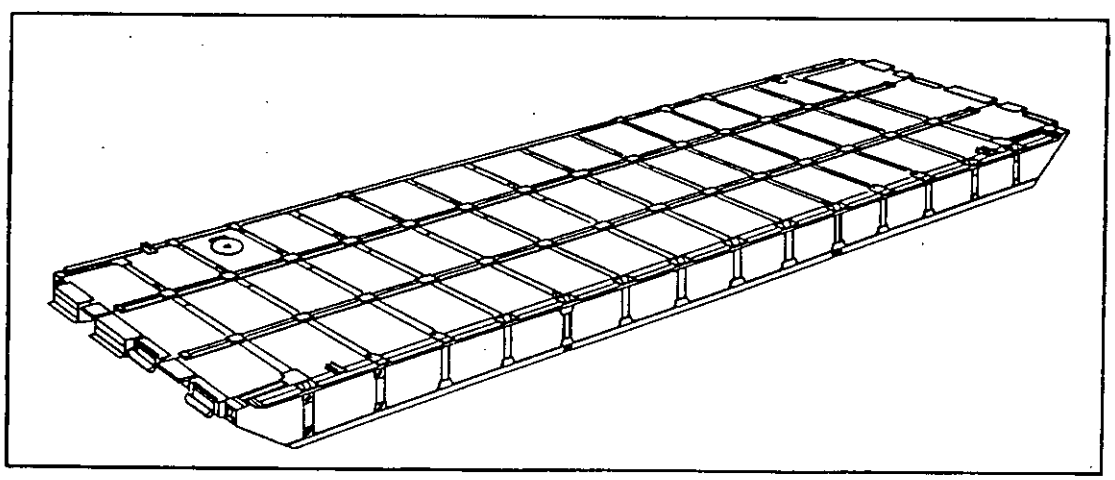


FIGURE 5
Harbor Lighter

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The pontoon yard craft have a variety of configurations and uses. A standard P-series pontoon is approximately 5 ft long x 7 ft wide x 5 ft deep (1.525 m x 2.135 m x 1.525 m) and can be connected with other pontoons to form structures that can serve many purposes such as causeway ferries, floating causeways, elevated piers and cargo lighters. The standard Navy Lighterage (NL) causeway section, shown in Figure 5, is constructed from standard pontoons and measures 90 ft long x 21 ft wide x 5 ft deep (27.45 m x 6.405 m x 1.525 m). It can be outfitted with end connectors and spud wells. These units are maintained in inventory at Naval Amphibious Construction Battalion Centers.

Container-size modular pontoons (Approximately 40 ft x 8 ft x 5 ft) (12.2 m x 2.44 m x 1.525 m) are currently under development for use in the above descriptions and transportation by containership.

2.3 Applicable Standards and Certifications. The primary standard for the design of all USN craft is Naval Sea Systems Command (NAVSEA) S9AA0-AA-SPN-010/GEN-SPEC, General Specifications for Ships of the United States Navy, 1985 Edition (22).

2.3.1 Organization. This document is organized in accordance with the NAVSEA Ship Work Breakdown Structural (SWBS) and is a comprehensive framework that can be used through the entire ship life cycle. It covers the major areas of all ship design, including:

a) general guidance and administration,
b) hull structure,
c) propulsion plant,
d) electric plant,
e) auxiliary systems,
f) outfitting and furnishings,
g) integrated engineering, and
h) ship assembly and support services.

Because the yard craft will be supported by the Navy and yard craft systems must often interface with other Navy ships built to the General Specification given in (22) referenced above, following the GENSPEC will often produce the craft design with a low life-cycle cost while increasing the reliability and availability of the craft. The craft designer must determine if the use of alternate standards will produce a more satisfactory craft and then recommend the use of those standards to the end-user or acquisition organization for approval.

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NOTE:

Yard craft systems must often interface with shoreside and shipboard facilities which were constructed to standards other than the GENSPEC (for example, a generator barge may be required to supply 208-volt power to a shoreside messing facility). When such deviations from GENSPEC practice are identified, the craft designer must determine alternate standards with which to design the craft, and must determine whether such deviation will adversely affect other aspects of the craft, such as life-cycle cost, safety, or interoperability with other Navy ships. Approval for such deviation will be reviewed by the end-user or acquisition organization.

Where the yard craft will be operated within or transiting the jurisdiction of non-DOD agencies, the standards and regulations which those agencies promulgate must be taken into consideration. Such agencies include the U.S. Coast Guard (USCG), the U.S. Public Health Service, the Environmental Protection Agency, the Federal Communications Commission, state and local governments and foreign governments (where applicable). The craft designer must determine if there will be a conflict between these standards and must recommend waivers or other solutions for adjudication through the end-user or acquisition organization.

2.3.2 Other Standards. Other generally applicable and acceptable standards for construction of yard craft are:

a) U.S. Coast Guard Code of Federal Regulations, Title 33 (33 CFR) and Title 46 (CFR) (6).
b) American Bureau of Shipping (ABS), Rules for Building and Classing Steel Vessels, Aluminum Vessels, Reinforced Plastic Vessels, etc., as well as other Rules, Guides, and Requirements (8).
c) Institute of Electrical and Electronic Engineers (IEEE), Standard 45, Recommended Practice for Electric Installations on Shipboard (12).
d) American Petroleum Institute (API) Standards (29).
e) American Society of Mechanical Engineers (ASME) Codes (30).
f) American National Standards Institute (ANSI) Standards (31).
g) Society of Automotive Engineers (SAE) Standards (32).
h) American Society for Testing and Materials (ASTM) Standards (33).
i) Society of Naval Architects and Marine Engineers (SNAME) T&R Bulletins (34).
j) Maritime Administration (MarAd) Standard Specifications (35).

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Additionally, generally accepted Navy design practices are covered in Design Data Sheets developed by NAVSEA (23). Substitution of alternate design practices should be justified by the designer in terms of the design considerations of para. 2.4.

In general, no certification by civilian authorities will be required. However, the end-user may require such certification for insurance purposes, to assure compliance with non-DOD regulations or to allow repair by local commercial facilities which typically comply to civil standards. In such cases, the designer will refer to the required certification organization's construction standards and will account for the inspection and certification practices.

2.4 Yard Craft Design Process. The design of yard craft is essentially an iterative process taken through several levels as described below to develop a yard craft of compatible structural, outfitting, mechanical, and electrical systems.

Because the requirements for and the configuration of yard craft may vary widely from craft to craft, this handbook describes the design process rather than the details. Following the process will allow the designer to account for the performance and construction features which must be present in USN yard craft. The details of the design are left to the designer, with guidance provided to assure compliance with the necessary standards and certifications.

To best use the guidance provided by this handbook, the designer is encouraged to follow through the handbook format at each level of the craft design effort, checking to be sure that all trade-offs and conflicts have been adjudicated satisfactorily before proceeding to the next level. If this process is followed, all aspects of the craft design will be properly accounted for, the iterations of the design effort (and therefore its cost and schedule) will be minimized and the resulting craft will best satisfy the requirements established by the end-user.

2.4.1 Design Aspects. The following aspects of each major system and integrated craft design must be accounted for and optimized within the limits of the design effort:

a) Performance vs. requirements,
b) Compatibility with other Navy ships and craft,
c) Technical description and specifications,
d) Service margin,
e) System safety,
f) Personnel safety,
g) Manning requirements,
h) Reliability, availability and maintainability
i) Installation design (including maintenance, access, and clearances),

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j) Inspection and certification requirements,
k) Testing,
l) Logistic supportability (including training, parts, etc.), and
m) Life-cycle cost (including acquisition, installation, fuel, inspection, maintenance, insurance, etc.).

2.4.2 Mission Requirements. The design process must begin with a reasonable definition of the yard craft mission requirements agreed upon by NAVFAC and the designer. These requirements must meet the needs of the NAVFAC mission as well as practical design standards. Some mission requirements to be considered for yard craft are:

a) dead weight carrying capacity and stability,
b) cargo handling equipment,
c) quarters, space arrangements, outfitting,
d) towing speed,
e) auxiliary services and power (heating, ventilation, water, refrigeration, waste, lighting, communications, etc.)
f) type of service (ocean, coastwise, harbor, river, locks),
g) towing, mooring, berthing, drydocking, fittings,
h) access for inspection, maintenance and repair,
i) materials, and
j) construction, operation and maintenance costs.

Once reasonable mission requirements are set, concept design can begin.

2.4.3 Concept Design. Concept design translates mission requirements into principal yard craft characteristics. Design feasibility is established through a brief study of alternative designs. Critical design characteristics can be varied in a parametric analysis to establish the most effective design solutions. Block arrangement sketches are prepared to develop efficient location of essential yard craft components and to develop approximate hull proportions for several concepts. Critical design characteristics such as vessel length, beam, depth, draft, power requirements, weight and cost are developed for each concept to allow selection of the most appropriate concept for preliminary design development.

2.4.4 Preliminary Design. Preliminary design refines and establishes the major vessel characteristics affecting performance, cost and safety. The design is developed to include the following major documents defining the design:

a) hull lines and offsets drawing,
b) general arrangement drawing,
c) structural midship section and deck scantling drawings,
d) hydrostatic characteristics,
e) weight estimate report,
f) intact and damage stability study,
g) towing speed power estimate or model test report,

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h) engineering system diagrams for fresh and sea water, fire extinguishing, sewage, heating, ventilation, lighting, air conditioning, hydraulic, steam, and electrical distribution systems,
i) Power loads analysis,
j) auxiliary machinery arrangement,
k) material and equipment specifications, and
l) cost estimate.

2.4.5 Contract Design. Contract design yields a set of plans and specifications which form an integral part of the shipbuilding contract document. It encompasses one or more iterations of the design, thereby, further refining the preliminary design. This stage delineates more precisely such features as hull form, general arrangements, towing and stability characteristics, structural details, engineering systems and material specifications. Paramount is a weight and center of gravity estimate taking into account the location and weight of each major item on the vessel. The final general arrangement is developed, fixing the overall volumes and areas of cargo, machinery, stores, fuel oil, fresh water, living and utility spaces and their interrelationship, as well as their compatibility with other features such as cargo handling equipment, and machinery components.

The accompanying specifications delineate quality standards of hull and outfit and the anticipated performance of machinery and equipment. They describe the tests and trials in order that the vessel will be considered acceptable. Refer to (22) for applicable military specifications.

2.4.6 Detail Design. The final stage of vessel design is the development of detailed working plans. These plans are the installation and construction instructions to the ship fitters, welders, outfitters, metal workers, machinery vendors, pipefitters, etc. Up to this stage, the design has been passed from one engineering group to another. In this final stage, the transfer is from engineer to shipyard worker.

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2.5 Safety Analysis. A safety analysis in accordance with MIL-STD 882B, System Safety Program Requirements, shall be performed at appropriate stages of the design process, with the intent that the design be altered should a safety hazard be identified. The designer may be able to provide a solution entirely within the scope of the system or subsystem in which the potential hazard exists. However, should the hazard be one requiring redesign of other aspects of the craft, a safety review by the designer and safety engineer in conjunction with the end-user or craft acquisition organization will be required to adjudicate the resulting trade-off.

The safety analysis of each system shall identify the failure modes of that system and shall determine the effects of each failure mode upon the safety of personnel both onboard the craft and on the other craft or pierside facilities adjacent to the craft under design, especially if the craft is supplying a service to the alongside facility.

Additionally, the safety analysis shall identify conditions of craft operation which require special attention, such as fuel discharge, during which additional personnel, specially trained personnel, additional equipment, or special procedures will be required. The safety analysis shall also identify potential hazards which may occur during the craft construction and testing process, before delivery to the end-user or craft acquisition organization. These hazards shall be minimized if possible during the design process, but those remaining should be noted in the craft construction specifications.

2.6 Supportability Analysis. A supportability analysis shall be performed at several stages of the design process to ensure that the supportability requirements set forth by the end-user or craft acquisition organization will be met during the life of the craft. A supportability analysis consists of a review of the craft systems in light of the specific supportability benefits outside the required scope of the requirements. Those features shall be examined to see if their inclusion suits the purposes of the end-user.

Where shortcomings are noted in the ability of the Navy system to support the craft as designed, the designer shall review those shortcomings with the end-user or the craft acquisition organization to adjudicate the resulting trade-offs.

The supportability analysis shall include, but not be limited to the following aspects of the craft support:

a) life-cycle cost
b) maintenance concepts
c) support and Test equipment
d) supply support
e) transportation and handling
f) technical data
g) facilities and
h) personnel and training.

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Section 3: NAVAL ARCHITECTURE

3.1 Introduction. Naval architecture deals with the development of the utility craft hull and superstructure to meet mission requirements for space and volume, outfitting and cargo handling gear, stability, speed and towing characteristics, structural strength and integrity, material preservation, and constructability.

3.2 General Arrangements. General arrangements define spaces for all the required functions and equipment, properly coordinated for location and access. There are four consecutive steps that characterize general arrangement: allocation of main spaces, setting individual space boundaries, locating equipment and furnishings within boundaries, and providing interrelated access. The space boundaries are generally defined as: cargo spaces, machinery spaces, crew, passenger and associated spaces; tanks and miscellaneous.

The main boundaries must meet the following requirements:

a) watertight subdivision and integrity to survive specific damage conditions as defined in para. 3.4,
b) firezone subdivision,
c) structural integrity, and
d) adequate access.

All equipment and machinery shall be designed and installed to facilitate access for purposes of maintenance repair and ease of operation. For further guidance pertaining to general arrangements refer to (1), (22), (23), Society of Naval Architects and Marine Engineers (SNAME), Ship Design and Construction (2), and USCG C110.1, Recommended Practice for Electrical Installation on Shipboard (6).

3.3 Hull Form. Hull form is influenced primarily by cargo carrying requirements (described in para. 3.2, General Arrangements) and by resistance, strength, stability and towing requirements. The following sections provide guidance and recommended analysis approaches and scope for the latter areas.

Yard craft are generally modified boxes in form. Key characteristics include proportions, rake slope and radius, bilge and deck radii, bow and transom elevation and symmetry.

3.3.1 Proportions. The following hull proportions limits, given in terms of Length-to-Depth (L:D) and Beam-to-Depth (B:D), summarize past experience in yard craft design and operation. These limits consider the effects of hull form on resistance, strength and maneuvering and are service dependent.

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For Length-to-Depth use the following service dependent ratios:

a) Full seagoing -- not more than 14:1
b) Coastwise -- not more than 16:1
c) Sounds and bays and limited service -- not more than 18:1

For Beam-to-Depth use a ratio of no more than 6:1.

3.3.2 Other Form Factors

3.3.2.1 Headlog Elevation. Set above load waterline so that freeboard is maintained at design tow speed.

3.3.2.2 Rake Slope and Rake Radius. As illustrated in Figure 6, a long rake slope induces pounding in rough water and should be restricted to river craft. Use a large rake radius to minimize or eliminate the plane areas of the rake slope. For rough water operation, shorten the rake radius to minimize pounding and to stiffen the unsupported areas of rake plating between frames.

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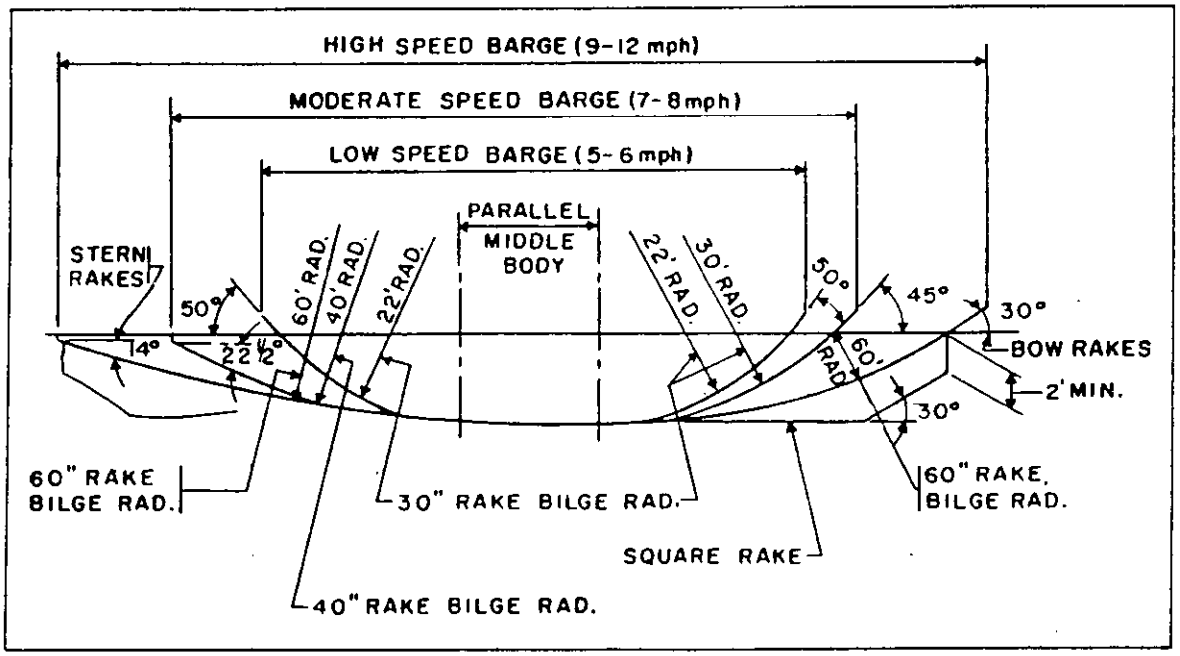


FIGURE 6
Recommended Rake Ends

3.3.2.3 Bilge Radius. A small bilge radius shall be incorporated in the design throughout the parallel midbody of the vessel. At the raked end, the bilge radius should be increased as much as practicable. The keel shall be flat.

3.3.2.4 Deck Lines. A small radius shall be incorporated in the design at deck corners. A flat deck without camber is preferred. Sheer may be incorporated in the foredeck if the craft is to be exposed to sea conditions. For simplicity of construction a flat sheer line is preferable to a curved line.

3.3.2.5 Skegs. Typical box-like hull forms that are usually found in nonself-propelled craft are notoriously apt to yaw back and forth under tow. Skegs are the usual remedy applied to cure this problem. Frequently the pair of skegs are constructed at an outboard angle to the craft's centerline to improve their effectiveness and may incorporate a small flat fin on their trailing edge. Yawing tendencies may also be improved by loading the craft with a trim down by the stern, towing with a bridle, rather than a single line, and by adjusting the length of the towline.

3.3.2.6 Maximum Towing Efficiency. Maximum towing efficiency compatible with a minimum of expensive and complicated structural shaping indicates a form rectangular in plan with round corners and simple raked bow and stern. Principle elements affecting towing resistance are primarily ratios of length-to-draft and length-to-beam, fairing of bow and stern, frontal area, and trim. Faired bows and sterns, while offering a maximum of towing efficiency, require complicated and expensive structural shaping. The square

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or blunt ended bow and stern can be readily constructed at minimum cost, but their towing characteristics are poor. Therefore, simple symmetrical raked bow and stern, that is, a moderate rise of bottom, while intermediate in efficiency and moderately efficient in ease of construction, appears to offer the most advantages with the least disadvantage. At the anticipated towing speed, the towing resistance would be approximately 1-1/2 times that of a faired type and one-half that of a box-type hull form.

3.4 Hydrostatics and Stability. The yard craft must have sufficient buoyancy and stability characteristics to carry the mission required loadings. Important supplemental guidance is provided in (1).

3.4.1 Weight Estimate. To ensure that stability requirements are met and to provide information for seakeeping analysis, a weight estimate for the yard craft shall be made. This estimate will be of a level of detail consistent with the design stage. The weight estimate will include the weight and the vertical, transverse and longitudinal centers of each weight item.

3.4.2 Hydrostatics. The hydrostatic properties of a vessel are required to calculate the load carrying capacity, stability and seakeeping characteristics of the vessel. These properties are generally presented in the form of a drawing called the "Curves of Form" or "Hydrostatic Curves" as well as in tabular format. These properties include:

DISPLACEMENT	-	displacement in long tons
DRAFT	-	height of waterline above keel in feet amidships
KB	-	height of center of buoyancy above keel
LCB	-	longitudinal center of buoyancy in feet from amidships (+fwd)
LCF	-	longitudinal center of flotation in feet from amidships (+fwd)
BML	-	height of longitudinal metacenter above center of buoyancy in feet
KML	-	height of longitudinal metacenter above keel in feet
BMT	-	height of transverse metacenter above center of buoyancy in feet
KMT	-	height of transverse metacenter above keel in feet
MT1	-	moment to change trim one inch in foot-tons
CIDOFTS	-	change in displacement for one foot trim by stern in tons per foot of trim
TPI	-	tons per inch immersion
VOLUME	-	displaced volume in cubic feet
WETTED SURFACE	-	surface area of wetted portion of hull in square feet
WPLANE AREA	-	area of waterplane in square feet
C_p	-	prismatic coefficient
C_b	-	block coefficient
C_m or C_x	-	midship section coefficient
VCG or KG	-	vertical center of gravity in feet (from keel)
GM	-	height of metacenter above VCG in feet

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3.4.3 Stability. Using the properties stated in para. 3.4.2, the following stability analyses may be performed as necessary:

3.4.3.1 Trim Lines. Calculation of the vessel trim condition after shell-to-shell flooding of vessel compartments.

3.4.3.2 Longitudinal Strength. Calculations of load, shear and bending moment along the length of a vessel in still water and with the vessel in a hogging or sagging condition on a wave. This information is a key input to subsequent structural design and analysis.

3.4.3.3 Floodable Length. Determination of the lengths of shell-to-shell compartments of specified permeabilities that when flooded will cause the ship to settle with the waterline tangent to the margin line, typically three inches below the freeboard deck.

3.4.3.4 Limiting Drafts. Determination of the maximum drafts fore and aft to which a vessel may be loaded prior to flooding and still survive when a specified group of compartments are damaged.

3.4.3.5 Intact Stability. Calculation of either curves of statical stability or cross curves. Statical stability curves should be calculated for still water and for a vessel balanced upon a wave if there is significant shape variation over the length of the hull. These curves indicate the value of righting arm available as the vessel heels through a range of angles.

3.4.3.6 Damaged Stability. Calculations made to determine drafts and stability when the vessel has sustained damage to one or more compartments.

3.4.3.7 Trim and Stability Report. Study of vessel stability for a range of load and damage conditions to establish satisfaction of stability criteria.

For many yard craft these properties and analyses are easily estimated by hand calculation, however for more complicated shapes and more accurate determination it is recommended that the Ship Hull Characteristics Program (SHCP), CASDAC #231072, developed by the Naval Ship Engineering Center be used. This program is widely available, inexpensive to use, and used extensively by naval and commercial interests. The SHCP is currently available through National Technical Information Services (NTIS).

Criteria defining acceptable limits for intact and damaged stability as well as description of stability test requirements can be found in (1), (6), (23) and NAVSHIPS 0901-882-0002, Technical Manual (26).

3.5 Seakeeping. Seakeeping comprises the response characteristics of a vessel in a seaway. The vessel motions (displacements, velocities and accelerations) in a seaway are heavily influenced by hull shape, form characteristics as determined above, vessel mass properties, tow speed and seaway energy content. These motions in turn define the survivability, operational ability and dynamic loads acting on hull and equipment.

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Seakeeping criteria applicable to yard craft fall into the following groups:

a) Displacement Limits (overturning, deck wetness)
b) Velocity Limits (slamming, structural loads)
c) Acceleration Limits (motion sickness, structural loads)

The level of seakeeping analysis appropriate is a function of vessel service area and vessel or cargo value or sensitivity to motion. The level of analysis can be limited to hand calculation estimates of natural periods, frequency domain motions analysis or, more rarely, include time consuming and expensive analysis such as model tests or time domain simulations.

3.5.1 Hull Form Characteristics. Two yard craft hull form characteristics, rake slope and rake radius, play a significant role in determining yard craft motions and resulting loads. A long rake slope induces pounding in rough water and should be restricted to river craft. Use of a large rake radius minimizes the plane areas of the rake slope and thus reduces slamming loads. A small bilge radius (or bilge keels) will reduce roll motions by increasing viscous damping (see Figure 6 for recommended combinations of rake slope and radii).

The key vessel mass characteristics are vessel displacement, mass moments of inertia and hydrodynamic properties such as added mass and damping which determine vessel response natural periods. Mass inertial properties can be determined from lightship weight estimate data, however, for most purposes they can be approximated by .40B, .25L, and .25L for roll, pitch and yaw inertial gyradii respectively where B is the vessel beam and L, the vessel length.

3.5.2 Estimates of Roll and Heave Responses. Estimates of roll and heave response periods can be made for preliminary evaluation using approximate formulae. The period of roll can be computed using Equation (1) or (2) as follows:

EQUATION: $$T_r = 1.108\ k\ /\ GM \qquad (1)$$

If the approximation, k=.40B is introduced, then,

EQUATION: $$T_r = 0.44\ B\ /\ GM \qquad (2)$$

where:

T_r = Roll natural period (seconds)
k = Mass radius of gyration of the vessel about it's roll axis
B = Beam (feet)
GM = Transverse metacentric height (feet)

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Roll motions often govern acceptable motion criteria. For preliminary evaluation, the angular acceleration, a_r, can be computed from:

EQUATION: $$a_r = 4(3.14)^2 \ (PHI)/57.3T^2 \quad (3)$$

where:

a_r = Angular acceleration (radians per second^2)
PHI = Angle of roll (degrees); typical value 20 degrees
T_r = Roll period (seconds)

Maximum athwartship loads will occur when the vessel is at the maximum roll amplitude. The tangential load is:

EQUATION: $$F = W\sin(PHI) + \frac{a_r}{g} Wr \quad (4)$$

where:

W = Weight of superstructure, cargo, or equipment (lb)
r = Radial distance from vessel KG to weight CG (ft)
g = Acceleration of gravity (ft/s^2)
KG = Vertical Center of gravity measured from keel (ft)

Similarly the heave natural period can be estimated very roughly using:

EQUATION: $$T_h = 9.26 \ DISP/Aw \quad (5)$$

where:

T_h = Heave natural period (seconds)
Aw = Waterplane area (ft^2)
DISP = Displacement (long tons)

Vertical heave accelerations normally can be considered to be less than 0.3g's for dynamic load applications. The vertical component of pitch acceleration can be as high as 0.75g's at the extreme ends of the vessel in moderate to heavy seas.

Further information on dynamic loads and vessel motions is given in (1).

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3.5.3 Vessel Hydrodynamic Responses. Vessel hydrodynamic responses also can be determined using a variety of computer codes. A recommended program is the Ship Motion Program (SMP), developed by the David W. Taylor Naval Ship Research and Development Center and is available through NTIS. Other available programs include RELMO, developed at the Naval Civil Engineering Laboratory (NCEL) and obtainable through NCEL, Code 51, and a series of seakeeping programs, (MITA, MITM, and MITQ) developed at the Massachusetts Institute of Technology and available through NAVSEA, Code 55W3. These codes predict vessel motions based on data defining vessel form, surface area, mass properties and seaway characteristics. Key seaway characteristics are stated in terms of energy content, frequency content, duration and approach direction. Seaways are often defined in terms of wave energy spectra or wave amplitude spectra. Several spectra representing various assumptions of seaway development, water depth and fetch are available. Some widely used ones are the Pierson-Moskowitz, Bretschneider, and JONSWAP spectra. Table 2 gives the relationship between sea state definition and Pierson-Moskowitz spectral parameters.

The recommended computer codes provide a reasonable prediction of all vessel motions including:

a) Amplitudes of surge, sway, heave, roll, pitch and yaw.

b) Velocities and accelerations in the six modes of motion -- valuable for dynamic load prediction.

c) Prediction of deck wetness and bow/stern emergence and slamming.

d) Motion sickness criteria -- usually expressed as a function of exposure time and magnitude of vertical acceleration. For short term exposure on yard craft for experienced personnel, a 0.2g working limit for significant acceleration is acceptable.

3.6 Towing. Hull form characteristics compatible with maximum towing efficiency have been described in para. 3.3.2, Other Form Factors. Towing performance can be greatly enhanced by selecting a hull form which minimizes towing resistance while providing acceptable directional stability and maneuverability characteristics. Directional stability can be enhanced through the attachment of appropriate skegs at the stern of the barge and through the use of a towing bridle. Vessel towing characteristics can be confirmed cost effectively by model testing. Additional valuable towing guidance is provided in NAVFAC MIL-HDBK-1038/5, Towing Nonself-Propelled Floating Structures (28), and in para. 3.8 of this handbook.

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TABLE 2
Definitions of Sea Conditions: Wave and Sea for Fully Arisen Sea

Sea state	Description	Wind				Sea								
		Beaufort wind force	Description	Range, knots	Wind velocity, knots †	Wave height, ft. Average	Wave height, ft. significant	Wave height, ft. Average 1/10 Highest	Significant range of periods, sec.	I_{max}, period of maximum energy of spectrum	$\bar{T}$ Average period sec.	L Average wavelength	Minimum fetch, nautical miles	Minimum Duration hr.
0	Sea like a mirror	0	Calm	Less than 1	0	0	0	0						
	Ripples with the appearance of scales are formed, but without foam crests.	1	Light airs	1-3	2	0.05	0.08	0.10	Up to 1.2	0.7	0.5	.10 in.	5	18 min.
1	Small wavelets, still short but more pronounced; crests have a glassy appearance but do not break.	2	Light breeze	4-6	5	0.18	0.29	0.37	0.4-2.8	2.0	1.4	6.9 ft	8	39 min.
2	Large wavelets, crests begin to break. Foam of glassy appearance. Perhaps scattered white horses.		Gentle breeze	7-10	8.5	.06	1.0	1.2	0.8-5.0	3.4	2.4	20	9.8	1.7 hr.
					10	0.88	1.4	1.8	1.0-6.0	4	2.9	27	10	2.4
3	Small waves, becoming larger; fairly frequent white horses.	4	Moderate breeze	11-16	12	1.4	2.2	2.8	1.0-7.0	4.8	3.4	40	18	3.8
					13.5	1.8	2.9	3.7	1.4-7.6	5.4	3.9	52	24	4.8
					14	2.0	3.3	4.2	1.5-7.8	5.6	4.0	59	28	5.2
					16	2.9	4.6	5/8	2.0-8.8	6.5	4.6	71	40	6.6

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TABLE 2 (Continued)
Definitions of Sea Conditions: Wave and Sea for Fully Arisen Sea

Sea state	Description	Wind				Sea								
		Beaufort wind force	Description	Range, knots	Wind velocity, knots †	Wave height, ft. Average	significant	Average 1/10 Highest	Significant range of periods, sec.	I_{max}, period of maximum energy of spectrum	$\bar{T}$ Average period sec.	L Average wavelength	Minimum fetch, nautical miles	Minimum Duration hr.
4	Moderate waves, taking a more pronounced long form; many white horses are formed (chances of some spray).	5	Fresh breeze	17-21	18	3.8	6.1	7.8	2.5-10.0	7.2	5.1	90	55	8.3
					19	4.3	6.9	8.7	2.8-10.6	7.7	5.4	99	65	9.2
5					20	5.0	8.0	10	3.0-11.1	8.1	5.7	111	75	10
6	Large waves begin to form; the white foam the crests are more extensive everywhere (probably some spray).	6	Strong breeze	22-27	22	6.4	10	13	3.4-12.2	8.9	6.3	134	100	12
					24	7.9	12	16	3.7-13.5	9.7	6.8	160	130	14
					24.5	8.2	13	17	3.8-13.6	9.9	7.0	164	140	15
					26	9.6	15	20	4.0-14.5	10.5	7.4	188	180	17
7	Sea heaps up and white foam from breaking waves begins to be blown in streaks along the direction of the wind (spindrift begins to be seen).	7	Moderate gale	28-33	28	11	18	23	4.5-15.5	11.3	7.9	212	230	20
					30	14	22	28	4.7-16.7	12.1	8.6	250	280	23
					30.5	14	23	29	4.8-17.0	12.4	8.7	258	290	24
					32	16	26	33	5.0-17.5	12.9	9.1	285	340	27
8	Moderately high waves of greater length; edges of crests break into spindrift. The foam is blown in well-marked streaks along the direction of the wind. Spray affects visibility.	8	Fresh gale	34-40	34	19	30	38	5.5-18.5	13.6	9.7	322	420	30
					36	21	35	44	5.8-19.7	10.3	10.3	363	500	34
					37	23	37	46.7	6-20.5	14.9	10.5	376	530	37
					38	25	40	50	6.2-20.8	15.4	10.7	392	600	38
					40	28	45	58	6.5-21.7	16.1	11.4	444	710	42

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TABLE 2 (Continued)
Definitions of Sea Conditions: Wave and Sea for Fully Arisen Sea

Sea state	Description	Wind				Sea								
		Beaufort wind force	Description	Range, knots	Wind velocity, knots †	Wave height, ft. Average	Wave height, ft. significant	Wave height, ft. Average 1/10 Highest	Significant range of periods, sec.	I_{max}, period of maximum energy of spectrum	$\bar{T}$ Average period sec.	L Average wavelength	Minimum fetch, nautical miles	Minimum Duration hr.
8	High waves. Dense streaks of foam along the direction of the wind. Sea begin to roll. Visibility affected.	9	Strong gale	41-47	42	31	50	64	7-23	17.0	12.0	492	830	47
					44	36	58	73	7-24.2	17.7	12.5	534	960	52
					46	40	64	81	7-25	18.6	13.1	590	1110	57
9	Very high waves with long overhangings crests. The resulting foam is in great patches and is blown in dense white streaks along the direction of the wind. On the whole, the surface of the sea takes a white appearance. The rolling of the sea becomes heavy and shocklike. Visibility is affected.	10	Whole gale *	48-55	48	44	71	90	7.5-26	19.4	13.8	650	1250	63
					50	49	78	99	7.5-27	20.2	14.3	700	1420	69
					51.5	52	83	106	8-28.2	20.8	14.7	736	1560	73
					52	54	87	110	8-28.5	21.0	14.8	750	1610	75
					54	59	95	121	8-29.5	21.8	15.4	810	1800	81

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TABLE 2 (Continued)
Definitions of Sea Conditions: Wave and Sea for Fully Arisen Sea

Sea state	Description	Wind				Sea								
		Beaufort wind force	Description	Range, knots	Wind velocity, knots †	Wave height, ft. Average	Wave height, ft. significant	Wave height, ft. Average 1/10 Highest	Significant range of periods, sec.	I_{max}, period of maximum energy of spectrum	$\bar{T}$ Average period sec.	L Average wavelength	Minimum fetch, nautical miles	Minimum Duration hr.
	Exceptionally high waves (small and medium-sized ships might for a long time be lost to view behind the wave). The sea	11	Storm*	56-63	56	64	103	130	8.5-31	22.6	16.3	910	2100	88
9	is completely covered with long white patches of foam lying along the direction of the wind. Everywhere the edges of the waves crests are blown into froth. Visibility affected.				59.5	73	116	148	10-32	24	17.0	985	2500	101
	Air filled with foam and spray. Sea completely white with driving spray; visibility very seriously affected.	12	Hurricane*	64-71	>64	>80‡	>128‡	>164‡	10-(35)	(26)	(18)	~	~	~

*For hurricane winds (and often whole gale and storm winds) required duration and fetches are rarely attained. Seas are therefore not fully arisen.

†A heavy box around this value means that the values tabulated are at the center of the Beaufort range.

‡For such high winds, the seas are confused. The wave crests blow off, and the water and the air mix.

Source: W. A. McEwen and A. H. Lewis, "Encyclopedia of Nautical Knowledge," p. 483, Cornell Maritime Press, Cambridge, MD., 1963. "Manual of Seamanship," pp. 717-718, vol. II, Admiralty, London, H. M. Stationary Office, 1952. Pierson Neumann, James, "Practical Methods of Observing and Forecasting Ocean Waves," New York University College of Engineering, 1953.

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3.7 Hull Structure. Yard craft structure must be designed to accommodate a wide range of applications and loading configurations as described in paras. 2.3 and 2.4.1. The following sections describe loadings to be considered when developing structural designs, design procedures, and standard structural materials and configurations currently in use. References (1) (2) and (8) are necessary design guidance when developing yard craft structure.

3.7.1 Loadings. The major loads imposed on sea-going structures are:

a) Primary stillwater and wave induced bending and shear from deadweight and vessel weight,

b) Dynamic loadings from vessel motions in a seaway including local accelerations as well as hull vibrations (refer to para. 3.5),

c) Local loads from cargo, equipment, mooring, docking, grounding, collision, etc., and

d) Thermal loads.

A combination of the most severe loads is usually selected as the nominal design load for a given structure. Refs. (1) and (2) provide guidance for establishing the appropriate loading schemes. Table 3 is a summary of some typical loading values but should be supplemented with data from these two references.

3.7.2 Design Procedures. Hull and superstructure scantlings can be developed for typical barge configurations, refer to (8). Separate publications are available for steel barges for offshore service and for steel vessels for service on rivers and intracoastal waterways. These publications provide formuli and tables for selecting typical structural configurations and components based on barge size, cargo space, and superstructure geometry as well as environmental loading conditions.

Vessel structure can also be developed from basic principles of structural analysis and design. The vessel hull is considered as a beam acted upon by primary loadings. A midship section modulus is developed from the longitudinally continuous hull structure. The shear forces and bending moments determined from hydrostatic calculations described in para. 3.4, are applied to determine stresses and an acceptable structural configuration.

Local structure is developed to withstand both the applicable primary stress at its location as well as the local loading.

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TABLE 3
Design Loadings

TYPE OF CRAFT	LOAD	INCLUDE IN CALCULATIONS
All types	Dead	Weight of all parts of vessel, machinery, stores, fixed equipment and its contents.
	Live	Areas of human occupancy--100 psf Storage area--250 psf Roof--40 psf
	Freeboard	Full load causing minimum freeboard: Seagoing craft--min. freeboard 5 ft (1.525 m) River craft--min. freeboard 1 ft/0.305 m
	Impact Factors	30 percent (moving loads)
	Wind	20 psf on superstructure only
	Hull	Towing, mooring, hydrostatic, and maneuvering including impacts. Derive loads from drag and resistance data refer to DM-38.5 and capacity of fittings and mechanisms to set up the required pull. Watertight Bulkheads: a) Bulkheads forming boundaries of tanks--hydrostatic heads = depth + 4 ft (1.22 m) b) Other bulkheads--hydrostatic head (test load) = height + 2 ft (0.61 m) Decks: Local machinery or equipment loads
	Special	Grounding and collision forces--cannot be quantitatively fixed; provided for by cant frames, collision bulkheads and breast hooks.
Seagoing craft	Hull[1]	Waves--use L/10 and L/20 trochoid waves as standard.
	Super-structure[1]	Dynamic load--vessel rolling through 40° (20° each side of vertical) in a seaway, and quartering in a seaway.

[1]Loadings in addition to applicable loads under "All Types."

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Computer-aided structural analysis procedures such as NASTRAN and MAESTRO are also available to the NAVY through NAVSEA, Code 55Y. The procedures depend upon developing a Finite Element Model (FEM) describing the essential structure of the vessel. All primary and local loadings are applied to this model and combined loading effects and resulting stresses considered. MAESTRO provides a particularly valuable structural optimization capability.

3.7.3 Construction Details. Typical construction details are illustrated in Figures 1 through 4 and Figure 6 as well as in (2). The design shall also adhere to the following:

a) For steel craft the steel thicknesses shall exceed the following minimums:

(1) Shell, deck, and bulkhead plating -- 5/16 in. (7.937 mm)
(2) Plates other than above -- 1/4 in. (6.35 mm)
(3) Rolled shapes -- 5/16 in. flange, 1/4 in. web.
(4) A corrosion allowance of 1/16 in. (1.587 mm) shall be added to all plating in contact with seawater

b) Transverse frames shall be used for most efficient type of framing including drydocking requirements.

c) Headlog shall be fabricated from 3/4-in. (19.05 mm) plating, flanged over both deck and rake, and reinforced with 1/2- or 5/8-in. (12.7 mm or 15.875 mm) stiffener angles. Support these parts with a heavy vertical channel extending from rake frame to deck beam.

d) The corner detail is an extension of headlog construction. Omit flanging at intersection of rake bilge plate and corner plate and substitute butt welded heavy bent corner plate, 1-1/2 to 2-1/4 in. (38.1 mm to 57.15 mm) thick.

e) Serrated sections may be used for plating stiffeners. Do not use serrated sections in dump scows where snagging is possible when buckets are used for loading or unloading.

f) Grounding frames (intermediate members between rake frames) shall be used from collision bulkhead to first or second transverse rake truss bottom chord. For rough water service, extend grounding frames from a short distance aft of the fore peak bulkhead to the headlog.

3.7.4 Materials

3.7.4.1 Structural Steels. Steels are the most widely used material for yard craft construction. Advantages include relatively high strength to weight ratio, weldability, and naval and industry experience in its use. Disadvantages include its high maintenance cost, due to corrosion which must be minimized by adequate protective coatings and maintenance measures. Refer to (2) for specifications of recommended steels.

3.7.4.2 Aluminum. Certain specific alloys of aluminum are approved for use in naval yard craft. Its lightweight and corrosion resistance are attractive, but it is usually more costly than the other materials listed. Refer to (2) for specifications of allowable alloys.

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3.7.4.3 Composites (including glass fiber, carbon fiber, etc.). These materials are being used in barges and in transportable boats as well as for pleasure craft. Although it generally is not as durable as metals it is an approved material for use in naval yard craft. Refer to Handbook of Composites (7) and (8) for recommended practice in fiberglass construction.

3.7.4.4 Timber. May be considered for construction of small craft for temporary use. Timber floating structures are subject to marine borer action, which can be reduced by chemical means. Caulking to maintain watertightness is a recurring maintenance problem. Refer to NAVFAC DM-2 Series, Structural Engineering, (28) for recommended practice in use of timber in marine construction.

3.7.4.5 Concrete. Prestressed and/or precast construction and ferrocement have been used successfully. The low strength to weight ratio of concrete requires a heavier structure to obtain desired strength characteristics. Concrete is also subject to spalling and cracking requiring maintenance. Allowable stresses shall be used for concrete and reinforcement as specified in DM-2 Series, except as modified by project requirements. Refer to NAVSHIPS 0982-019-1010, Ferro-cement Boat Building Manual (25).

3.8 Equipment and Fittings. The following is guidance for some essential yard craft equipment and deck fittings. Refer to (2) for more detailed guidance.

3.8.1 Line Equipment. Cleats, chocks, fairleads and bitts of adequate size and strength shall be provided. Reinforce structure locally to provide for line pulls resisted by these fittings.

For yard craft normally anchored in open water, anchor fittings, billboard for lashing anchors, and deck winch/windlasses shall be provided.

3.8.2 Fendering System. For pontoons provide upper and lower fenders with verticals all around pontoon.

For equipment barges provide system similar to that of floating cranes and derricks (refer to DM-38.1 (28)).

3.8.3 Access Fittings

3.8.3.1 Hatches and manholes. Each compartment shall be provided with a hatch or minimum size manhole, 16 x 27 in. (406.4 mm x 685.8 mm). Two accesses or escape hatches are required for all compartments which are normally occupied by the crew. Hatches that normally remain open, and those to fuel/fresh water compartments, shall be raised above deck. Other hatches and manholes shall be the flush type.

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3.8.3.2 Companionways. Companionway ladders, doors (Watertight [WT], and Non-watertight [NWT]) shall be provided for efficient movement of personnel throughout craft.

3.8.4 Bridles. A towing bridle, of width approximately equal to the beam of the vessel, with each leg length 1-1/2 times the beam shall be provided. Locate towing lugs close to headlog to reduce chafing. Refer to MIL-HDBK-1038/5 (28) for further guidance on towing gear.

3.8.5 Other Fittings and Equipment. Yard craft requiring operating crews shall be outfitted with handrails, grab rails, and grab bars. Use portable handrails where advantageous. Provide specified quantities of ring buoys, life rafts, kapok vests, and related stowage facilities.

Navigational lights shall be provided in accordance with USCG Publication No. CG-169, Rules to Prevent Collisions of Vessels.

Provide tank vents and piping in accordance with (22).

For open sea towing, a removable or hinged steel wave bulwark generally is provided bow and stern. Their shapes, heights, and strengths should be adequate to deflect waves upward or laterally, thus reducing the quantity of free water which would flow over the deck.

3.8.6 Machinery and General Storage Compartments. Machinery and general storage compartments shall be located away from the deckload area. Raised watertight hatches or manholes, ladders, and companionways shall be provided for easy access and ventilation. Use removable antislip steel flooring within the compartment.

3.8.7 Fuel and Fresh Water Compartments. The design of fuel and fresh water compartments shall include:

a) Fill pipes, valves, and vents (flame screened for fuel),
b) Raised manholes for hatches and access, and
c) Ladders under the manholes.

3.9 Vessel Preservation. Vessel preservation is dependent upon developing a design with few built-in maintenance problems. The designer should consider ease of maintenance of the structure and systems by providing easy access for maintenance. Structural details should not create maintenance problems, such as poor weld details or zones which collect water. Materials shall be selected to prevent corrosion and painted surfaces should be properly prepared. Standard preservation procedures for paint or cathodic protection are given in (2).

3.10 Test Plans and Certificates. Both the USN and the USCG have the requirement that a stability test be conducted and a report prepared to demonstrate that the vessel meets stability criteria stated in (6) and (26). These tests must be conducted after construction or major overhauls to

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the vessel. Reference (2) lists the numerous tests and inspections of structural, mechanical and electrical system required after new construction or major overhaul. Dock and at sea-trials are conducted to complete testing and confirm vessel speed and power characteristics.

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SECTION 4: MARINE ENGINEERING

4.1 General. This section describes the marine engineering aspects of yard craft design. Marine engineering encompasses the primary electrical, mechanical, and fluid power systems of the yard craft. It also includes the necessary auxiliary systems such as drainage, firefighting, and ventilation which support the primary systems.

4.1.1 Major Systems. The following major systems are described in paras. 4.2 through 4.22:

a) Liquid Cargo Storage and Transfer,
b) Electrical Generation and Distribution,
c) Steam Generation and Distribution,
d) Mechanical Power Production and Transmission,
e) Diesel Engine Support,
f) Boiler Support,
g) Hydraulic Power,
h) Compressed Air Production and Service,
i) Sea Water Service,
j) Fresh Water Production and Service,
k) Fire Extinguishing,
l) Drainage and Ballast,
m) Heating, Ventilation and Air Conditioning,
n) Incineration and Solid Waste Disposal,
o) Sewage Collection and Plumbing Drains,
p) Environmental Pollution Control,
q) Deck Drains,
r) Refrigeration,
s) Lighting,
t) Communications, and
u) Special Systems (e.g., tank cleaning, vacuum, etc.).

In general, yard craft are too small to allow the designer to consider each electrical, mechanical and fluid system independently of the others. An optimum craft design integrates many systems and minimizes the installed equipment, fuel requirements, and logistic support requirements.

4.1.2 Other Design Considerations. Careful specification of equipment, piping, wiring, fastening and fluid materials will increase the value of the design effort and the craft. Refer to the USN GENSPEC for guidance in the selection of materials. Reference (22) refers to military standards and specifications that establish the allowed materials for many items. Where no guidance is given or a choice of several materials is left to the designer, the designer must justify his choice of materials in terms of the design considerations of para. 2.3.

Additionally, consideration shall be given to corrosion, mechanical damage, fluid contamination, chemical resistance, static electricity, and field repairability.

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The installation design, generally developed in the detail stage of craft design, determines a significant part of the initial and life-cycle cost of the craft. Consideration of factors such as skid-mounting or modularization of assemblies, access openings and routes, inspection and maintenance clearances, number of take-down pipe joints, etc., will affect the value of the craft.

4.2 Major Yard Craft Additional Design Guidance: Liquid Cargo Storage and Transfer.

4.2.1 Liquid Cargo Storage. Storage tanks shall be provided and arranged to reduce the possibility of contamination of liquid cargo by seawater or by other onboard liquids. If the liquid is flammable or hazardous, storage shall be provided in accordance with the applicable USCG regulations. Access to the tanks shall be provided for inspection and cleaning. Provision shall be made for sounding all tanks with minimum effort.

4.2.2 Liquid Cargo Pumps. Liquid cargo pumps and their prime movers shall be selected to be capable of delivering the required flow at the required pressure for the complete range of cargo viscosities over the expected craft operational temperature range. Should a cargo heating system be required, it shall be capable of providing the necessary heat from onboard sources alone. Pressure relief valves shall be provided to protect pumps and piping from overpressure damage. Pump controls shall be located near cargo valve operating stations to minimize manning requirements. Refer to (22) and (35) for pump design and installation requirements.

4.2.3 Liquid Cargo Transfer System. The liquid cargo transfer system shall be arranged for ease of operation to reduce manning requirements. Piping shall be designed to minimize pressure losses to keep pumping power requirements and therefore craft power generating requirements to a minimum.

4.3 Major Yard Craft Additional Design Guidance: Electrical Generation and Distribution

4.3.1 General Requirements. For the selection of equipment and the layout of the electrical system, refer to (22) and (12).

4.3.2 Nature of Supply. Power shall be generated and distributed at 480 VAC, 3-phase, 60 Hz, unless the craft is primarily intended for use in supplying electric power to a shoreside or shipboard distribution system which requires other supply characteristics. Low voltage loads and lighting branch circuits shall be supplied with 120 V, single-phase power via transformers.

4.3.3 Shore Power Connection. If required to have a shore power connection, the craft shall be provided with weather boxes that have covers capable of excluding spray and dirt from the connectors. One box shall be provided on each side of the craft. An interlock shall be provided at the main switchboard to prevent paralleling shore power with onboard generators.

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4.3.4 Generators. Generators shall be of the brushless type and if located in wet areas shall be suitably protected from rain and spray. Where possible, generators, switch gear, and mechanical/fluid support machinery shall be located as near to each other as possible to reduce piping/wiring runs and to reduce operating and maintenance manning. All generators shall be of the same type and capacity.

4.3.5 Generator Capacity. The generator capacity shall be selected after conducting a load analysis. Follow the guidance provided by NAVSEA Design Data Sheet 310-1 (23), with the exception that high load factors shall be used for special onboard equipment, such as cargo pumping, as these will primarily determine the system capacity required. The power required to be supplied to a shoreside facility or other ship shall be included with a load factor of 1.0. The load analysis shall examine all major operating conditions under extremes of heat and cold to identify the maximum power requirements. Onboard generating capacity shall be sufficient to produce the required maximum load with all but one generator operating, while the operating generators shall be loaded to no more than 90 percent of their continuous maximum capacity.

4.3.6 Emergency Power. In general, emergency power shall be provided for navigation lights, communications, exit lights, and critical control equipment. Unless the load proves to be too large or the required time too long, emergency power shall be provided by rechargeable batteries. Battery chargers shall be provided and shall be connected to the craft's power distribution system. Should an emergency generator be required, it shall be sized to supply sufficient power to all emergency loads connected to it simultaneously, including off-craft loads identified as emergency loads.

4.3.7 Navigation Lights. Navigation lights shall be powered by rechargeable batteries. Batteries shall have sufficient capacity to allow operation of the navigation lights without recharging during the maximum towing period identified by the craft design requirements. Photo-electric switches shall be provided to turn off the lights during daylight conditions. Navigation lights shall be provided with automatic lamp-changing mechanisms to allow unattended operation for the extent of the maximum towing period identified.

4.3.8 Grounding and Bonding. Particular care shall be taken in designing the grounding and bonding the system of the craft. Because of the possibility that the yard craft will be electrically connected to a ship or shoreside facility with a damaged electrical system, provision shall be made for protection against stray currents through the craft hull. If the craft is a platform for welding operations, the potential for hull damage is even greater.

Equipment grounding connections shall be led to the minimum practicable number of hull grounding points to minimize stray currents through the hull. Refer to Sections 21.4 and 21.5 of (12)) and MIL-STD-1310, Shipboard Bonding, Grounding, and other Techniques for Electromagnetic Comparability and Safety, for further details pertaining to grounding and bonding of electrical systems.

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4.3.9 Lighting. Low level red illumination shall be provided for darkened ship conditions.

4.4 Major Yard Craft Additional Design Guidance: Steam Generation and Distribution

4.4.1 General Requirements. For guidance in the design of the steam generation and distribution system refer to (22).

4.4.2 Boilers. Boilers shall be marine-type package units including all support equipment and controls. Boilers shall be arranged for unattended, failsafe operation and fired with NATO F-76 fuel oil.

4.4.3 Water Treatment. Water treatment equipment shall be provided to allow use of potable water from shoreside distribution systems. If the craft is required to operate independent of shoreside water sources, equipment shall be provided to produce boiler-quality water from the water in which the craft will operate.

NOTE:

This may include harbor water, which is often contaminated with oils and chemicals not found in the open ocean.

4.4.4 Steam Generating Capacity. A steam load analysis, similar to the electric load analysis described elsewhere in this design manual, shall be performed to determine the maximum steam load. Heat balances shall be performed according to SNAME T&R Bulletin 3-11 (34) for each load condition. The combined boiler capacity shall be sufficient to produce the maximum steam load with no boiler operating at more than 90 percent of its continuous rated capacity. The boiler feedwater system and makeup system shall be sized to match 100 percent of the boiler continuous rated capacity.

4.4.5 Steam-Driven Machinery. All major machinery shall be electrically-powered rather than steam-driven. This will reduce the load on and size of the boiler. For example, all onboard pumps shall be electric-motor driven. Where possible, electric heat shall be used rather than steam heat.

NOTE:

An exception is cargo heating, where electric heating may be a safety hazard. Minor items, such as pilot-operated steam system regulating valves, may be steam-driven if practicable.

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4.4.6 Shore Connection. If a shore connection for steam is required, it shall be provided with the following (in order, from outboard to inboard):

a) cap which seals the shore connection,

b) flange and piping designed to withstand the maximum working pressure of the shoreside branch to which it will be connected,

c) shutoff valve on the main deck at the connection flange rated to the maximum working pressure of the shoreside branch to which the craft will be connected,

d) strainer with overboard dirt blowdown,

e) water trap with automatic dump,

f) pressure-reducing station which will lower the incoming steam pressure to the working pressure of the craft steam system, and

g) pressure relief valve sized to relieve steam pressure above the maximum working pressure of the craft steam system.

4.5 Mechanical Power Production and Transmission

4.5.1 Prime Movers. Prime movers shall be diesel engines, coupled to the consumer of mechanical power by a suitable transmission system (refer to paras. 4.5.2 through 4.5.5). The fuel shall be No. 2 diesel, also known as DF-2, NATO F-54, or VV-F-800C. In addition, Naval Distillate Fuel (NDF), NATO F-76, or MIL-F-16884 and Jet Fuel, JP-5, NATO F-44, or MIL-T-5624 shall be considered as interchangeable for reliable operation.

NOTE:

Diesel engines operating on jet fuels are less tolerable of dirt and water, therefore, the fuel should be water-free and filtration should be at least down to the 5 micron level. Due to variations in the heating values of substitute fuels, minor throttle adjustments will be necessary to obtain the desired power output. Where the engines will be run in other than flat sea conditions, they shall be installed with their crankshaft axis parallel to the craft's longitudinal axis.

Where they will be run only in sheltered waters, they may be oriented to suit the available space. Engines shall be suitable for marine service and shall be protected from sea spray and rain. To prevent personnel injury, exposed moving parts, such as shafts, fans and belts, shall be shielded by appropriate guards in accordance with American National Standards Institute (ANSI), ANSI Standards, B15.1 (5).

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4.5.2 Clutches. Clutches shall be provided to allow the diesel engines to be started and run without operating the driven equipment. Friction clutches shall be suitably enclosed to prevent loss of friction from spray or rain. Remotely operated clutches shall be arranged for failsafe operation in the event of failure of their electric, hydraulic, pneumatic or mechanical control power source.

4.5.3 Gear Boxes. Where change of prime mover speed or direction is required to drive equipment, a gear box rated for continuous use shall be provided. The gear box shall be suitably enclosed and lubricated. Where required, the gear box lube oil shall be cooled by sea water or air through a heat exchanger.

4.5.4 Shafting. Shafting shall be sized in accordance with guidance given in (22) and (8). Shaft material and coatings shall be chosen to avoid corrosion.

4.5.5 Controls. Diesel engines shall be arranged for local starting and stopping, and for remote stopping at appropriate locations. Appropriate locations include the control station for fueling operations, etc.

4.6 Diesel Engine Support

4.6.1 Fuel Storage, Transfer and Service. Fuel storage tanks shall be equipped with:

a) a fill connection with closure (cap or valve),
b) a suction connection (may be combined with fill connection),
c) a vent to weather with gooseneck, closure, ball-check valve and flame screen,
d) a sounding tube with closure,
e) a drain or docking plug, and
f) appropriate linkage for bonding/grounding during fuel transfer.

If there is a separate fuel service (day) tank or if the fuel storage tank also serves as a service tank, the following shall also be provided:

a) a visual level gauge with stop valves,
b) an engine supply connection with quick-closing valve, and
c) an engine return connection with quick-closing valve.

4.6.1.1 Fuel Tank Vent - Diesel. The fuel tank vent shall be sufficiently high to avoid taking water into the tank during moderate towing and operating conditions, and shall not be located near a source of ignition. A coaming or other enclosure shall be provided around the fuel tank vent and the fill connection to prevent discharge of fuel into the water. The visual level gauge shall be located to avoid mechanical damage.

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4.6.1.2 Fuel Service Tanks. The fuel service tank shall be located to allow the engine-mounted fuel service pump to take suction from a nearly empty tank. A priming pump or transfer pump shall be provided to allow priming of a fuel service pump if the fuel service tank is not sufficiently high to allow gravity filling of the system.

Fuel connections to the engine shall be equipped with approved flexible sections to allow for engine deflection. A stop valve shall be provided on the craft side of each flexible section.

An engine-driven fuel service pump shall be provided. A fuel strainer shall be provided on the suction side of the service pump.

4.6.2 Engine Air Intake and Combustion Exhaust. Combustion air for the diesel engines shall be taken from the space in which they are installed, or from the weather if installed on deck. Suitable ventilation shall be provided in each engine compartment to prevent engine air starvation. Suitable protection shall be provided to prevent aspiration of rain, spray, sand, or foreign objects into the engine air intake.

Combustion exhausts shall be led to weather as far as practicable from personnel and from ventilation intakes. Additional care shall be taken to keep engine exhaust piping from being a source of ignition at pierside facilities or alongside other ships. Exhaust piping shall be insulated to prevent maximum surface temperature from rising above 125° F (51.6° C), and the insulation shall be suitably protected from physical damage as well as wetting. Asbestos insulation material is not to be used.

4.6.3 Lube Oil Storage, Transfer and Service. The diesel engine lube oil service system shall be self-contained within the engine package. The lube oil sump shall be large enough to allow unattended operation for 24 hours. The system shall be configured to provide proper lubrication at all static and dynamic angles expected to be encountered when the engine is in operation.

The lube oil service pump shall be engine-driven. An engine sump drain with a locked closed valve shall be provided. A sump fill cap shall be provided which positively closes the fill connection. Where the engine is in a location exposed to the weather, the sump fill cap shall be spring closing.

For engines with lube oil sump capacity less than 5 gal. (18.925 L), a rack shall be provided to hold cans of make-up lube oil. For engines with larger sumps, a storage tank shall be provided with a locking oil can fill valve, sight glass, and vent which is protected from spray.

4.6.4 Engine Cooling. Engines shall be cooled by a closed fresh water system. The fresh water pump shall be engine-driven. Where the fresh water is cooled by air passing over a radiator, the radiator fan shall be engine-driven and enclosed in a shroud to protect personnel. The installation of a radiator cooled engine shall be such as to provide a continuous supply of fresh air, and to direct the heated discharge air away from the fan intake.

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Where the fresh water is cooled by sea water passing through a heat exchanger, the sea water pump shall be engine-driven. Where the sea water pump suction will not be continuously flooded by external sea water, a positive displacement pump shall be provided. The fresh water system shall be protected from freezing by the use of anti-freeze chemicals. The sea water system shall be provided with drains at the low points inboard of the hull valves to allow draining of the piping when the craft is inactive during freezing weather.

4.6.5 Cold Weather Starting. Suitable means shall be provided to allow rapid starting and load pick-up in cold weather without damage to the engine. These means shall include (as recommended and approved by the engine manufacturer) jacket water and lube oil preheaters, fuel preheaters or air preheaters. Glow plugs and booster fuel ("ether") shall not be provided.

4.7 Boiler Support

4.7.1 Fuel Storage, Transfer and Service. Fuel storage tanks shall be equipped with:

a) a fill connection with closure (cap or valve),
b) a suction connection (may be combined with fill connection),
c) a vent to weather with gooseneck, closure, ball-check valve and flame screen,
d) a sounding tube with closure,
e) a drain or docking plug, and
f) appropriate linkage for bonding/grounding during fuel transfer.

If there is a separate fuel service (day) tank or if the fuel storage tank also serves as a service tank, the following items shall also be provided:

a) Visual level gauge with stop valves.
b) Engine supply connection with quick-closing valve.
c) Engine return connection with quick-closing valve.

4.7.1.1 Fuel Tank Vent-Boiler. The fuel tank vent shall be sufficiently high to avoid taking water into the tank during moderate towing and operating conditions, and shall not be located near a source of ignition. A coaming or other enclosure shall be provided around the fuel tank vent and the fill connection to prevent discharge of fuel into the water. The visual level gauge shall be located to avoid mechanical damage.

4.7.1.2 Fuel Service Tank-Boiler. The fuel service tank shall be located to allow the boiler-package- mounted fuel service pump to take suction from a nearly empty tank. A priming pump or transfer pump shall be provided to allow priming of the fuel service pump if the fuel service tank is not sufficiently high to allow gravity filling of the system.

Fuel connections to the boiler shall be equipped with approved flexible sections to allow for thermal growth deflection. A stop valve shall be provided on the craft side of each flexible section.

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A boiler-package-mounted fuel service pump shall be provided. The pump shall be electrically driven. A strainer shall be provided on the suction side of the pump.

4.7.2 Boiler Air Intake and Combustion Exhaust. Combustion air for the boiler shall be taken from the space in which it is installed, or from the weather if installed on deck. Suitable ventilation shall be provided in each boiler compartment to prevent boiler air starvation. Suitable protection shall be provided to prevent aspiration of rain, spray, sand, or foreign objects into the boiler air intake.

Combustion exhausts shall be led to weather as far as practicable from personnel and from ventilation intakes. Additional care shall be taken to keep boiler exhaust piping from being a source of ignition at pierside facilities or alongside other ships. Exhaust piping shall be insulated to prevent maximum surface temperature from rising above 125° F (51.6° C), and the insulation shall be suitably protected from physical damage as well as wetting. Asbestos insulation material is not to be used.

4.7.3 Feedwater. Feedwater shall be provided in accordance with the boiler manufacturer's recommendations and approval. A make-up feedwater tank shall be provided and installed to protect the contents from contamination. The fill connection shall be labeled "Boiler Feedwater Only", and shall have a locked closed cap. The tank shall be provided with:

a) Fill connection (as above),
b) Feed pump suction connection,
c) Vent to compartment with gooseneck, closure, ball-check valve (if vented to weather),
d) drain valve,
e) visual level gauge with stop valves, and
f) sampling connection with valve.

4.7.4 Cold Weather Starting. Suitable means shall be provided to allow safe starting of the boiler in cold weather (when the boiler is cold). These means shall include, as recommended and approved by the boiler manufacturer, fuel preheaters, feedwater heaters, air preheaters, or boiler water circulators.

4.7.5 Controls. The boiler shall be arranged for local starting and shutdown, and for remote shutdown at appropriate remote locations. Appropriate locations include the control station for fueling operations, etc. The boiler shall be arranged for automatic, unattended operation. The ABS rules and guidelines for unattended auxiliary boilers as well as the USCG regulations covering unattended auxiliary boilers are applicable.

4.8 Hydraulic Power

4.8.1 Hydraulic Power System. Where hydraulic power is required to operate a piece of equipment, such as a crane or windlass, a dedicated hydraulic power supply shall be provided to each consumer, or to groups of

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consumers with similar functions if located near each other. Each hydraulic power system shall be a closed loop system. Where a critical craft function has been identified, two independent hydraulic power supplies shall be provided to that consumer. Each hydraulic power supply shall be complete with reservoir, pump(s), control valves, strainers, filters, and fluid cooler. The maximum system working pressure shall be 3,000 psi. Where practicable, the hydraulic power supply shall be located interior to the craft; however, if it is installed in the weather, all components shall be of suitable materials and coatings, and shall be protected from spray.

4.8.2 Pumps. Hydraulic power pumps shall be of a positive displacement type and driven by either an electric motor or diesel engine.

4.8.3 Piping. Rigid piping shall be provided for the hydraulic power system of materials and thickness to suit the maximum system design pressure. Flexible hose shall be provided where the piping connects to hydraulic pumps, motors, rams, and moving apparatus. Relief valves shall be provided to protect piping and components from overpressure. Sufficient slack shall be left in hoses and sufficient U-sections shall be provided in rigid piping to allow for contraction at peak pressures.

4.9 Compressed Air Production and Service

4.9.1 Compressors. Compressors shall be belt-driven by electric motors. Compressors shall be air-cooled and oil-lubricated by a self-contained lubricating system. Compressors shall be fitted with unloading valves to reduce the starting load on the motor. They also shall be fitted with intake air filters.

4.9.2 Receivers. Receivers shall be mounted to allow accumulation of condensed moisture at the drain connection and far from the compressed air discharge connection. Separate receivers shall be provided for diesel engine starting air.

4.9.3 Piping. Drains shall be located at points where condensed moisture may accumulate. Onboard service outlets shall be fitted with a shutoff valve. If the craft compressed air system is to provide compressed air to a shoreside facility or alongside ship compressed air system, a non-return valve shall be installed at the craft deck discharge station to prevent entry of contaminated air into the craft system. Pressure-reducing stations shall be provided to reduce receiver pressure to 35 psig for service air.

4.10 Sea Water Service

4.10.1. Pumps. A minimum of two pumps shall be provided for sea water service. Each pump shall be capable of producing the required fire-fighting flow and pressure. Sea water service pumps, if used for fire-fighting, shall not be arranged to be used as bilge pumps or in off-craft pumping service.

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4.10.2 Piping. Suitable materials shall be used for sea water piping to minimize corrosion. Pipes shall be sized to maintain flow velocity between 6 and 12 ft/sec (1.83 and 3.66 m/sec). For additional guidance, refer to Section 505 of (22) and the Standard Spec for USN Craft. Drains shall be provided at the low point of all piping located in unheated compartments or on deck to allow draining of piping to prevent damage from freezing.

4.10.3 Saltwater Flushing. Provide hose connections on deck to allow flushing of deck areas with sea water.

4.10.4 Fire-Fighting. Provide fire hose stations on deck such that no more than 100 ft (30.5 m) of hose are required for coverage of the entire craft. Provide a deck connection and valves such that pierside pressurized fire-fighting water may be supplied to the craft's fire hose stations without inter-connection to the sea water service system.

4.11 Fresh Water Production and Service

4.11.1 General. Fresh water systems, whether for potable water, wash water, or boiler feed, shall be arranged strictly in accordance with Section 532 of (22).

4.11.2 Storage. At least two potable water storage tanks shall be provided if onboard berthing is provided.

4.11.3 Production. Fresh water production capabilities shall be provided if the craft will be used in areas where fresh water of sufficient quality or quantity is not available. The total production capability shall not be less than the anticipated maximum average daily consumption. At least two independent watermakers shall be provided, with the total required capability divided between them. Where potable water will be produced, disinfection equipment shall be provided.

4.11.4 Pumps. For potable water systems, at least two pressure pumps shall be provided, each providing half the anticipated maximum system flow, and arranged in a pressure lead-lag manner. The pumps shall feed a potable water pressure tank. The pumps shall be arranged to be self priming, or to have a continuously flooded suction.

4.11.5 Shore Connection. Where a connection to shore or to another ship is required, a backflow preventer and pressure regulator shall be installed to prevent damage to downstream piping or contamination to the upstream supply.

4.12 Fire Extinguishing

4.12.1 Agents. For enclosed machinery spaces, Halon-type agents shall be provided. For berthing, commissary, office, and utility spaces, as well as open machinery spaces, portable chemical or CO_2 agents shall be provided. Where munitions, torpedo fuel, or other flammable chemicals will be present on

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the craft, suitable agents and systems such as AFFF type foam or water spray shall be provided. Where Halon systems are installed, provisions are required to insure prompt evacuation of personnel, in accordance with para. 1.6 of NFPA 12A (10) and 12B (11).

4.12.2 Fixed System. A fixed, Halon-type system shall be provided for each enclosed machinery space. The storage bottles and actuating mechanisms shall be protected and located outside the machinery space. A fixed chemical system, if provided, shall be installed for galley equipment.

4.12.3 Portable Equipment. Portable fire extinguishers of the appropriate chemical, CO_2 or foam type shall be installed in several locations around the craft, both on deck and interior.

4.13 Drainage and Ballast

4.13.1 Pumps. Drainage and ballasting/deballasting functions may be combined and performed by the same pumps. Where the spaces served by a drainage pump are liable to contain oily water or hazardous chemicals, that pump shall not also be used for firemain service. Two independent means shall be provided for drainage/deballasting of each space. One shall be a pump driven by electric motor or by a diesel engine; the second may be a similar pump or may be an eductor powered by the firemain or other sufficient and reliable source of seawater. The pumps shall be constructed of materials suitable for seawater service. Where deep well or submersible pumps are provided, they shall be suitable for continuous immersion and operation in seawater. Ballasting of dedicated clean ballast tanks may be accomplished by the use of flood valves with locked-closed operators located on the main deck and protected from mechanical damage. However, these valves shall not exceed 10-in. (254 mm) Nominal Pipe Size, (NPS), and no more than one valve maybe installed for each compartment to be flooded. A remote shutdown station shall be provided on the craft weather deck near the overboard discharge if the pump is capable of draining a compartment likely to contain oily water or a hazardous chemical.

4.13.2 Piping. Piping shall be provided to allow ballast tanks to be deballasted from a "pressed-up" condition in two hours. The overboard discharge for all drainage and deballasting systems shall be located above the highest load waterline, and shall be provided with a screw-down non-return valve operable from the weather deck to prevent ingress of sea water if the discharge should be submerged in an extreme ballasting condition.

4.14 Heating, Ventilation and Air Conditioning

4.14.1 Machinery Space Ventilation. Ventilation shall be provided in occupied machinery spaces such that the difference between exhaust air temperature and supply air temperature shall not exceed 30° F (-1.1° C). All machinery spaces shall be positively ventilated by at least one exhaust blower, and preferably one supply blower in addition, and shall be designed to maintain the space under a slight negative pressure to prevent escape of vapors into adjacent compartments. Weather air intakes and exhausts shall be

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arranged to minimized recirculation of air, intake of combustion exhaust, intake of flammable vapors when alongside a pier or other ship. Sufficient forced or natural supply air shall be available in spaces containing diesel engines or boilers to allow proper combustion air intake.

4.14.2 Storage Space Ventilation. Mechanical ventilation shall be provided for all storage spaces containing flammable or hazardous fluids, chemicals, or bottled gases. It is generally recommended that mechanical ventilation for flammable storage areas provide six air changes per hour. Other storage spaces may be naturally ventilated unless otherwise required. Covered holds into which men may occasionally be required to enter shall be provided with a means to ventilate them. This may be a portable blower attached to appropriate deck fittings.

4.14.3 Berthing Space Ventilation. Berthing spaces and other occupied spaces shall be ventilated as described in Section 512 of (22).

4.14.4 Heating. Heating of occupied compartments, machinery spaces, and other spaces which require heating due to their use or their contents shall be accomplished by electric heat or by steam/hot water produced by an onboard boiler. A trade-off study shall be performed to determine the most suitable source of heat, depending on the requirements for electric power and steam for other services. Preference shall be given to electric heat, as it is more probable that pierside electric power will be available than steam. Where steam heat is provided, it shall be a two-pipe system which is closed within the occupied spaces. Thermostatic controls shall be provided within each occupied space to prevent overheating. Where unit air conditioners are provided, the heating function may be provided by those units. Consider explosion hazards when choosing the heat source.

4.14.5 Air Conditioning. Air conditioning shall be provided for all berthing spaces and other non-machinery spaces required to be continuously occupied. Unit air conditioners, such as "window"-type or overhead-type, may be used. However, the compartment served must still be provided with fresh air ventilation.

4.15 Solid Waste Disposal

4.15.1 General. Provision shall be made for collection and holding of solid waste, including food waste, as local ordinances may not permit overboard discharge of solid waste disposal.

4.15.2 Food Waste. The discharge drains of food grinders and dishwashing machines shall be piped to the sewage holding tank, or to a separate solid waste holding tank.

4.16 Sewage Collection and Plumbing Drains

4.16.1 Sewage Collection. A vacuum sewage collection system shall be provided where onboard toilets are required. A sewage holding tank (which may be part of the collection system) shall be provided of sufficient size to

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hold the maximum quantity of sewage anticipated in 24 hours. A USCG-approved marine sanitation device may be provided where local ordinances allow the overboard discharge from such a device. A transfer pump shall be provided to empty the holding tank.

4.16.2 Plumbing Drains. Plumbing drains from berthing and commissary spaces shall be led to a grey water holding tank which may not be part of the sewage collection, holding and transfer system. A transfer pump shall be provided to empty the holding tank. Deck drains shall be led overboard through non-return valves.

4.16.3 Piping. Pipe materials shall be suitable for sewage service. Threaded connections shall not be used. Where possible, sewage collection plumbing drain piping shall not pass through commissary, food storage, or ventilation machinery spaces. Where this cannot be practically avoided, no take-down joints shall be installed within those spaces.

4.16.4 Shore Connection. Shore connections in appropriate locations shall be provided for emptying the sewage holding tank and the grey water holding tank into shoreside facilities, using the onboard transfer pumps. Additionally, a bypass shall be provided around each transfer pump such that a shoreside pump may take suction on the tanks.

4.17 Environmental Pollution Control

4.17.1 Discharge of Oil. Suitable means shall be provided to prevent the overboard discharge of oil. A coaming with a drain port which may be plugged shall be provided around each fuel tank vent and fill connection. A drip pan or coaming shall be provided around each diesel engine, hydraulic motor, or other machinery item which may leak oil onto a weather deck.

4.17.2 Discharge of Sewage Waste. Refer to para. 4.15.

4.17.3 Discharge of Industrial Waste. Where the yard craft is used to store, transfer, or process materials which are prohibited from overboard discharge, suitable means shall be provided to contain any material which leaks. Additionally, any waste product from an onboard industrial process (e.g., degreasers, paint, welding slag) shall be contained onboard for transfer ashore.

4.18 Deck Drains

4.18.1 Piping. Weather decks above the main deck shall be provided with a drip coaming and drain pipes which spill onto the main deck. Weather areas of the main deck shall drain overboard through freeing ports, if bulwarks are provided, or directly off the deck edge. Interior decks shall drain overboard through non-return valves. Interior decks below the highest load waterline, which are required to be drained, shall drain into a holding tank with an automatic transfer pump.

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4.18.2 Coamings. All coamings and drip pans required for spill containment shall be fitted with drain plugs interior to the coaming and a nipple exterior to the coaming such that a collection device, pump or drip can, may be temporarily attached.

4.19 Refrigeration

4.19.1 Refrigerated Storage. Refrigerated storage shall be provided in accordance with applicable sanitation requirements. Where quantities to be stored are less than 200 ft^3 (6 m^3), multiple, self-contained, commercial refrigerator/freezers shall be provided. Air-cooled units shall be properly ventilated.

4.19.2 Refrigeration Machinery. Where built-in refrigerated storage is required with separate refrigerating machinery, a dual-machinery arrangement shall be provided for each space. There shall be no interconnection between the dual units. Each of the dual units shall be of sufficient capacity to hold the storage space at the required temperature. The total capacity of the dual units, working simultaneously, shall be sufficient to reduce the temperature in the space from ambient to the required freezer or refrigeration temperature (refrigerant R-12 shall be used).

4.20 Lighting

4.20.1 Interior Lighting. Interior lighting shall be provided in accordance with Illuminating Engineering Society recommendations (refer to (9)). Where occupied interior spaces are provided, emergency lighting shall be provided in the event the normal electrical supply fails. For continuously lit spaces, fluorescent lighting shall be provided. For unoccupied areas, such as storage spaces, incandescent lighting shall be provided. Explosion-proof fixtures shall be installed where there is the possibility of ignition of flammable vapors. All fixtures shall be protected from mechanical damage by use of suitable guards. Suitable retaining devices shall be provided to prevent lamps from loosening due to craft motion or vibration.

4.20.2 Deck Lighting. General deck lighting shall be provided in accordance with Illuminating Engineering Society recommendations. General deck lighting shall be provided by incandescent lamps. Where industrial or special-purpose work will be performed on deck, additional lighting shall be provided by high-pressure discharge lamps, switched independently from the general lights. Emergency lighting shall be provided at gangways and other means of egress from the craft.

4.20.3 Navigation Lights. Navigation lights shall be provided in accordance with USCG regulations of (3). Where the craft will be towed offshore and unmanned, suitable automatic lamp-changing fixtures with batteries shall be provided.

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4.21 Communications

4.21.1 Interior Communications System. Interior communications circuits shall be provided between occupied machinery and utility spaces and appropriate deck and berthing locations. These circuits shall be independent of the normal source of electrical power and operated by both a normal and emergency (generator or battery) power source. Interior communication circuits shall allow for the sounding of a general alarm as well as point-to-point voice communication.

4.21.2 Exterior Communications Equipment. Fixed exterior communications equipment shall be provided where the craft will be towed while manned. Additionally, where the craft is manned and is supplying a critical service to a ship or shoreside facility, a fixed exterior communications system shall be provided. The antennas shall be located to prevent contact with personnel or with machinery such as cranes.

4.22 Special Systems. Where special-purpose electrical, mechanical or fluid systems are required, the design of their installation shall generally follow that of the conventional systems. Examples of special systems are tank-cleaning systems, vacuum solids transfer systems, fixed welding systems (gas and electric), and chemical storage and transfer systems.

4.22.1 Safety Hazards/Pollution Abatement. Where specific safety hazards or pollution abatement considerations are identified with these systems, the design of the installation shall consider the reduction of safety hazards and the reduction of potential pollution as first priorities. Control of the special system shall be centralized in order to reduce the manning requirements for the craft. Shut-down stations for tank cleaning and similar systems shall be located on deck at one or more readily accessible locations. Where possible, the system shall be configured to fail-safe.

4.23 Test Plan. A test plan for individual systems and subsystems shall be developed in the design stage. The test plan shall include a description of the test to be performed, the test equipment required, the safety precautions necessary during testing, the stage of construction at which the test is to be performed, the data to be recorded, the method of evaluation of the data, and criteria for passing and failure.

4.24 Test Development. For descriptions of tests pertaining to specific craft systems refer to Section 095 of (22).

The number of inspections and tests of a given component or installation shall be kept to a minimum consistent with the required results. Unless otherwise specified, all arrangements for handling and supporting weights (including weights of personnel), all arrangements for tackling heavy strains, and all parts upon which the safety of the craft or of life depend, shall be given a static load test equal to twice the specified working load. In the case where

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the working load is not specified, the test load shall be based on the expected duty of the machinery, component, or fitting. For hoisting arrangements, the static test load shall be suspended clear of all supports and held suspended for a sufficient period to permit inspection of welds and other fastenings, but need not be lifted or moved by the equipment under test. After relieving the static load, there shall be no evidence of permanent deformation of structure. Weight handling arrangements shall be tested to demonstrate capacity or handle a load equal to 1-1/2 times the working load at reduced speed.

4.24.1 System Operational Tests. Wherever possible, tests on machinery shall be performed in conjunction with related system operational tests. Electrically operated machinery shall be tested with power from the ship service generators or shore power at the specified voltage, phase and frequency. Air and hydraulically operated machinery shall be tested to demonstrate satisfactory operation over the specified range.

Should more than one function be driven by the same electric or hydraulic motor, the test shall be repeated for each function. Conversely, if more than one power unit drives a given function, the test shall be repeated for each.

4.24.2 Recording Test Data. During tests of machinery, electrical and fluid systems, the following minimum data, as applicable, shall be recorded:

a) Convenience of control,
b) Duration of test,
c) Electric motor input parameters,
d) Pressures,
e) Safety devices,
f) Speed for driven auxiliary,
g) Stroke of variable stroke pumps,
h) Temperatures of the compartment (ambient) and injection seawater, and the temperature of working fluids for each test condition at the beginning and completion of the test, and,
i) The time or speed of performing various operations (such as hoisting, etc.).

During tests of electric machinery, the ability of electric brakes to stop and hold under the specified conditions shall be proven by a sudden interruption of the power supply. During testing, foundations and hold-down bolts and nuts shall be examined to detect signs of inadequacy.

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APPENDIX A

Conversion of Commonly Used Measurements[1]

1/16 inch	=	1.6 mm
1/4 inch	=	6.5 mm
1/2 inch	=	13 mm
5/8 inch	=	16 mm
5/16 inch	=	8 mm
1] inch	=	38 mm
2] inch	=	57 mm
15 inches	=	380 mm
23 inches	=	580 mm
36 inches	=	900 mm
1 ft	=	0.3 m
5 ft	=	1.5 m
30 ft	=	9 m
80 ft	=	24 m
100 ft	=	30 m
70 tons	=	63.5 metric tons
254 tons	=	230 metric tons
0.60 cu ft	=	0.017 cubic meter
2.80 cu ft	=	0.079 cubic meter
50 psig	=	0.35 MPa
100 psig	=	0.69 MPa
40 psf	=	195 kg/m^2
100 psf	=	488 kg/m^2
250 psf	=	1220 kg/m^2
-20° F	=	-28° C
0° F	=	-18° C
33° F	=	0.5 C
100° F	=	38° C

[1]Conversions are approximate.

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Military Documents are available from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. Telephone: Autovon (DOD only): 442-3321; Commercial:

Naval Sea Systems Command (NAVSEA) Documents

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(24) 0938-LP-018-0010 Heating, Ventilation, and Air Conditioning Design Criteria Manual for Surface Ships

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(26) NAVSHIPS 0901-882-0002 Technical Manual, Chapter 988C

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DM-2 Series	Structural Engineering
DM-3 Series	Mechanical Engineering
DM-4 Series	Electrical Engineering
DM-38.01	Weight Handling Equipment
DM-38.02	Dredging Equipment
MIL-HDBK-1038/5	Towing Nonself-propelled Floating Structures
P-401	Pontoon System Manual

Department of Defense activities may obtain copies of Design Manuals and P-Publications from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. Department of Defense activities must use the Military Standard Requisitioning and Issue Procedure (MILSTRIP), using the stock control number obtained from NAVSUP Publication 2002.

Other Government Agencies and commercial organizations may procure Design Manuals and P-Publications from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(29) American Petroleum Institute (API) Standards, 1220 L St. N.W., Washington, D.C. 20005

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MILITARY STANDARDS:

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(37)	MIL-STD-882E	System Safety Program Requirements

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